Yi Fan (Eric) Wang

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Education

University of Toronto (St. George) Toronto, ON, Canada, Sep. 2022 – Jun. 2027 (Expected)

- Honours Bachelor of Science
- Data Science Specialist (ASIP), Statistics Major, and Computer Science Minor
- **Academic Performance**: A+ in Data Science II, Databases, and Time Series Analysis; A in Machine Learning and Deep Learning
- **Relevant Courses**: Machine Learning, Deep Learning, Time Series Analysis, Databases, Data Science, Methods of Data Analysis, Survey Design and Data Preprocessing, Probability and Statistics

Research Experience

Undergraduate Research Student, University of Toronto Toronto, ON, Canada, Sep. 2026 – Present

- Member of a three-student undergraduate team on "Correlation-Augmented Neural Retrieval - From Hard Filtering to Spectral Diffusion," supervised by Professor Nick Koudas (Department of Computer Science)
- Project tests whether adding a sparse graph of corpus term-association estimates to a dense neural retriever can reduce off-topic results, recover relevant passages that share no query terms, and control query cost
- Team plans to compare four ways to inject that signal: candidate pre-filtering, metric warping for re-scoring, multi-hop graph diffusion, and contrastive fine-tuning, with an evaluation protocol and an assumption audit

Work Experience

Data Operations Engineer (Co-op), AISHU Technology Corp. Shanghai, China, May – Aug. 2025

- Built Text-to-SQL analytics across 3 customer POCs and 3 distinct large databases, answering target questions in approximately 1–13 minutes where manual analysis was impractical; personally demonstrated both delivered POCs
- Held majority responsibility in a 3-person team for business-rule design and SQL evaluation; personally designed approximately 400 queries across the first 2 POCs, covering approximately 10 and 20 question types
- Ran 9 DeepSeek V3 business-rule and prompt-tuning rounds (3 per POC); the first 2 POCs each improved Text-to-SQL execution accuracy from about 50% to about 70%
- Translated ambiguous requirements through 5 formal customer consultations and substantial follow-up into 30+ Markdown rule documents totaling approximately 6,000+ chunked rows across all 3 POCs

Logistics Assistant/Data Analyst, Marubeni-Itochu Steel Canada Inc. Toronto, ON, Canada, May–Aug. 2024

- Collected, cleaned, and analyzed 5 years of Canadian steel import data across 10 product categories and global trade partners using Excel and Statistics Canada datasets
- Compared pre-, during-, and post-COVID net weight patterns across categories and 5+ major partner countries, identifying shifts used in a 5-year Canada import trend analysis
- Created charts and summaries for internal stakeholders and maintained structured datasets on regulatory changes, supporting data-driven planning and reporting
- Supported logistics operations by validating invoices and shipping documents in GSAP, tracking vessel status, and assisting monthly reconciliations across 15+ warehouses

Projects

My Second Brain by YFW Jun. 2026 – Present

- Sole developer of a local-first LLM knowledge application that compiles text, document, image, and web sources into a provenance-linked, Obsidian-compatible wiki with hierarchical scopes and vision transcription
- Integrated scope-aware hybrid retrieval using BM25 and vector search with Reciprocal Rank Fusion, incremental reconciliation of changed/deleted Markdown, optional LLM reranking, and BM25 fallback on vector failures
- Shipped v1.0.0 as a Python 3.11+ package and CLI with a responsive FastAPI/TypeScript/Vite web app, OpenAI/Anthropic adapters, and Pydantic-validated outputs; verified by 176 passing Python tests and a 7-of-7 deterministic citation-grounding evaluation

Global CO2 Per-Capita Emissions Analysis with World Bank API Apr. 2026

- Independently built a 7,378-row country-year panel for 1990–2023 by retrieving and merging 6 World Bank API indicators; after ISO3 joins, aggregate-region filtering, log transforms, and centering year at 1990, retained 5,499 complete cases spanning 196 countries
- Modeled emissions drivers with OLS and GAM in Python (pandas, statsmodels, pygam), then extended to a pruned Decision Tree and tuned XGBoost using 5-fold country-grouped randomized search with cross-validated early stopping and 70/15/15 train/validation/test splits (136/30/30 countries) to prevent country-level leakage
- Tuned XGBoost achieved test R^2 of 0.722 (RMSE 0.543, MAE 0.359), outperforming the pruned Decision Tree (R^2 0.706); gain-based importance ranked log GDP, renewable-energy share, and urbanization highest. Published a reproducible Quarto/GitHub Pages site with animated maps, decade-level distributions, and GDP-emissions views across 3 interactive Plotly visualizations

LLM Source Classification with Survey and Text Features (Python) Nov. 2025

- Collaboratively developed a 3-class classifier to distinguish ChatGPT, Claude, and Gemini responses using a balanced 825-observation dataset of survey features and free-text responses
- Personally inspected and cleaned raw data, standardized labels, transformed Likert and multi-select fields, constructed the unified feature matrix with capped-vocabulary TF–IDF text features, and applied consistent preprocessing across data splits
- Compared bagged trees, softmax regression, and a tuned feed-forward neural network; the FFNN led on the held-out test set with 69% accuracy and 0.69 macro F1

NLP Classification: Tweet Sentiment and Text Emotion (Python) Apr. 2025

- Jointly built NLP pipelines for 3-class sentiment classification of COVID-19 tweets and 6-class emotion classification of 18,000 labeled short-text samples
- Compared count and TF–IDF representations with multinomial Naive Bayes and softmax regression, using 5-fold stratified cross-validation and PCA tuning
- Selected a 2,000-component count/PCA/softmax model with 84.0% test accuracy and 83.53% mean validation accuracy

Global Life Expectancy Regression Analysis (R) Dec. 2024

- Jointly analyzed how health, economic, education, and development indicators relate to global life expectancy using a 1,649-observation WHO dataset in R
- Built and refined multiple linear regression models with variable transformations, VIF-based multicollinearity checks, partial F-tests, and AIC-driven model selection
- Used diagnostics and ggplot2 visualizations to evaluate model assumptions and identify key predictors including schooling, GDP, adult mortality, and income composition of resources

Meal Master (Java) Dec. 2024

- Proposed the initial concept and helped assemble a five-person team to build a modular Java recipe-management application with recipe search, favorites, user accounts, and shopping lists, integrating TheMealDB and File.io APIs
- Owned the core implementation of the UserDataAccessObject and RecipeDataAccessObject and held primary responsibility for the core Shopping List use case; collaborated on Signup, Login, Logout, and Recipe Search use cases
- Coordinated regular meetings and progress tracking, informally coordinated integration work, and performed detailed code reviews covering correctness, tests, architecture, style, and documentation

Certifications

Supervised Machine Learning: Regression and Classification DeepLearning.AI, Stanford University

Generative AI for Everyone DeepLearning.AI

LangChain for LLM Application Development DeepLearning.AI

Building Systems with the ChatGPT API DeepLearning.AI

Skills & Interests

- **Software and Programming**: Python, Java, R, SQL, C, Git, Quarto, MS Office Suite
- **Coding Agents**: Claude Code, Codex, Cursor
- **Data Science**: pandas, NumPy, SciPy, scikit-learn, XGBoost, TensorFlow, statsmodels, pygam, PyTorch, Hugging Face Transformers, Sentence Transformers, Model2Vec, Faiss, hnswlib, NLTK, LangChain
- **Visualization**: Plotly, Folium, plotnine, Tableau, Matplotlib
- **Languages**: English, Mandarin
- **Interests**: Swimming, Piano, Chinese history